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Optimal Steering Control of Long Combination Vehicle with Multiple Steered Units

Minimizing Swept Path Width, Off-Tracking and Rearward Amplification

Master's thesis in Mobility Engineering

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Abstract

The introduction of Long Combination Vehicles (LCVs) with enhanced load-carrying capacities is a significant advancement in reducing carbon emissions and improving logistical efficiency. By decreasing the number of vehicles on the road, these LCVs contribute to a more sustainable and streamlined freight transport system. However, the increased length and load of these vehicles present challenges in maneuverability in tight spaces and lateral stability at high speeds. This research explores innovative solutions, such as multiple steered and propelled axles, which, when optimally controlled, address these challenges and enhance LCVs performance.
The primary focus of this thesis is the lateral dynamics of LCVs, specifically examining lateral off-tracking, swept path width, and rearward amplification. We utilize an A-double long combination vehicle equipped with two steerable and propelled axles as actuators. To improve maneuverability and lateral stability, we develop mathematical expressions for the aforementioned performance characteristics. These expressions serve as cost functions, which are minimized to determine the optimal control inputs for our actuators.
The optimization process is conducted using the CasADi toolbox in MATLAB, which provides a robust framework for defining the system dynamics and constraints necessary for the optimization problem. We evaluate the performance of the cost functions across various maneuvers and actuator configurations, highlighting the benefits of multiple actuators with optimal control allocation. The results demonstrate significant improvements in maneuverability at low speeds and lateral stability at high speeds, underscoring the potential of advanced control strategies in enhancing LCVs performance.

Keywords: Long Combination Vehicle, Optimal Control Allocation, Swept Path Width, Lateral Off-Tracking, Rearward Amplification, .

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Additionally, we want to express our special thanks to our beloved friends and parents for their unwavering support and companionship. Their encouragement gave us the confidence to face the unknown and overcome difficulties, even in the most trying circumstances.

Kartik Shingade and Shreekara Ramesh, Gothenburg, August 2024

ϕ_1

v_1

Nomenclature

Below is the nomenclature of indices, sets, parameters, and variables that have been used throughout this thesis.

Indices

i	vehicle unit number
j	instance in manuever

Parameters

m_1	mass of tractor (in kg)
m_2	mass of first trailer (in kg)
m_3	mass of dolly (in kg)
m_4	mass of second trailer (in kg)

Variables

X	x coordinate (in m) of CoG of tractor in global reference
Y	y coordinate (in m) of CoG of tractor in global reference
δ_1	Steering angle of tractor (in radian)
δ_2	Steering angle of dolly (in radian)
v_{1x}	Longitudinal velocity of tractor (in m/sec)
v_{1y}	Lateral velocity of tractor (in m/sec)
ϕ_1	Yaw angle of tractor (in radian)
$\dot{\phi}_1$	Yaw rate of tractor (radians/sec)
v_{2x}	Longitudinal velocity of first trailer (in m/sec)
v_{2y}	Lateral velocity of first trailer (in m/sec)

ϕ_2	Yaw angle of first trailer (in radian)
$\dot{\phi}_2$	Yaw rate of first trailer (in radians/sec)
v_{3x}	Longitudinal velocity of dolly (in m/sec)
v_{3y}	Lateral velocity of dolly (in m/sec)
ϕ_3	Yaw angle of dolly (in radian)
$\dot{\phi}_3$	Yaw rate of dolly (in radians/sec)
v_{4x}	Longitudinal velocity of second trailer (in m/sec)
v_{4y}	Lateral velocity of second trailer (in m/sec)
ϕ_4	Yaw angle of second trailer (in radian)
$\dot{\phi}_4$	Yaw rate of second trailer (in radians/sec)

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1
Introduction

In this thesis, the control strategies for actuators in long combination vehicles (LCVs) are optimized to enhance maneuverability during tight corners and other demanding driving scenarios. This is crucial for ensuring safety and efficiency in the operation of these large, multi-trailer vehicles. To achieve this, two distinct vehicle models are employed: the kinematic model and the kinetic model.
The kinematic model simplifies the system by focusing on the geometric aspects of vehicle movement, disregarding forces and inertia. It is particularly useful for analyzing path tracking and low-speed maneuvers, where dynamic effects are minimal. On the other hand, the kinetic model or single track model incorporates the dynamics of the vehicle, including forces, torques, and the influence of mass distribution. This model is essential for understanding high-speed behaviors, stability, and the effects of external forces.
Both models are developed and analyzed in-depth to evaluate their respective advantages and limitations, using distinct cost functions based on the objective of optimization. Key performance metrics such as maneuverability, stability, and computational efficiency are considered. By simulating various driving scenarios, the models are compared and cost functions are evaluated to determine their effectiveness in real-time applications.
The comparative analysis aims to identify the most suitable model for practical implementation in LCVs. This involves assessing the trade-offs between the simplicity and computational efficiency of the kinematic model and the comprehensive, albeit more complex, kinetic model. The ultimate goal is to select a model that provides robust control while being feasible for real-time application, ensuring that LCVs can navigate tight corners and handle nervous situations with improved precision and safety.

1.1 Background

Transportation of goods is one of the key elements in maintaining the livelihood of any country's subjects and its economy. Road freight transport alone accounts for approximately 49.3% of all road transport in the European Union (EU) [1]. LCVs have the potential to significantly reduce CO2 emissions by approximately 24–38% [2] and operational costs by around 30–50%, compared to existing commercial vehicles. The increased capacity of LCVs compared to traditional commercial vehicles also suggests that, in the future, fewer LCVs (and their drivers) will be required to transport the same amount of cargo, potentially improving traffic fluidity on roads.

LCVs are vehicle combinations consisting of one prime-mover (truck or tractor) and multiple trailing units (such as trailers or semitrailers with dollies). One of the foremost challenges for LCVs are associated with their maneuverability at low speeds and stability at high speeds due to their increased length and number of articulations, which limits their rapid market introduction and penetration. However, the dolly presents an economically viable platform for addressing these challenges. This is mainly because it allows existing, unmodified semitrailers to remain in operation, which are the most frequently used trailing units in the EU, responsible for approximately 80–85% of cargo transportation [3].
The potential for dolly innovation can be comprehended from reducing swept path, off-tracking, rearward amplification and assist in power and reverse- parking. This paper specifically addresses the first 3 things by proposing an optimized dolly steering system using a single control structure and later adding propulsion on dolly. For demonstration purposes, a types of LCVs is considered: the A-double (i.e., tractor-semitrailer-dolly-semitrailer), which has the highest potential from an economic perspective.

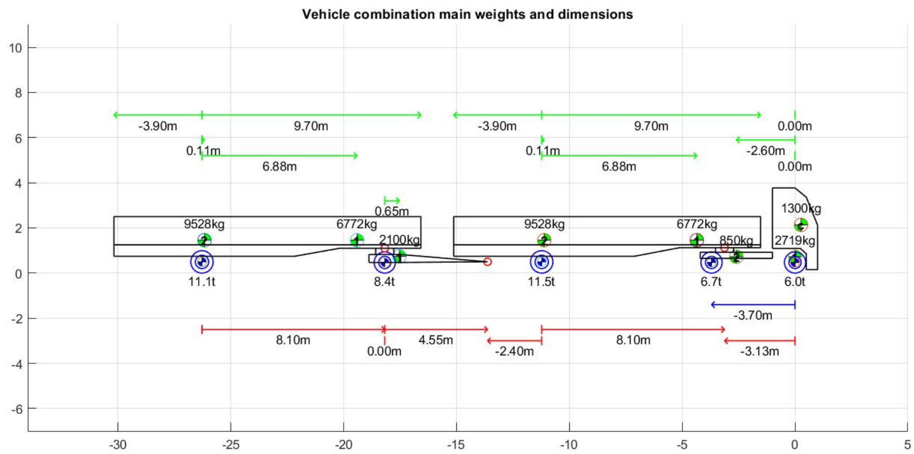


Figure 1.1: A-double combination configuration.

The Performance Based Standards (PBS) are used as measures for evaluating the performance enhancement of the considered LCVs due to the implementation of the steered dolly, compared to their conventional non-steered dolly versions. The PBS measures considered for low-speed maneuvers include swept path width and tail swing, while the PBS measures for high-speed performance include rearward amplification. To achieve this, both the most simplistic kinematic model and the single-track model are used for analysis. By comparing these models, the study aims

to identify the most effective control strategy for real-time application, ensuring enhanced maneuverability and stability of LCVs in various driving conditions.
The comprehensive analysis provided in this paper aims to demonstrate the benefits of optimized dolly steering and propulsion. It not only addresses the current limitations of LCVs in terms of maneuverability and stability but also paves the way for their broader adoption in the transport sector, ultimately contributing to environmental sustainability and operational efficiency.

1.2 Purpose

In this thesis, we aim to investigate the lateral dynamics of LCVs, focusing on the A-double configuration. The lateral dynamics characteristics of interest include the swept path width (SPW) at low speeds, rearward amplification (RWA), and lateral off-tracking at high speeds.

LCVs like the A-double, with steering and propulsion control primarily from the tractor, face challenges when maneuvering in tight spaces such as roundabouts and intersections. By adding additional steering and propulsion actuators to the trailing units (here dolly) of the LCV, these maneuverability challenges can be mitigated.

However, a new challenge arises in distributing control signals to these actuators optimally to achieve the desired objectives, whether maneuvering in tight spaces or preserving lateral stability at high speeds.

To address this, we employ an optimal control allocation approach to distribute control signals to the actuators on the LCV, aiming to achieve specific objectives. These objectives are formulated as mathematical cost functions, and by evaluating these cost functions, we determine the optimal control allocation.

To this end, we developed both kinematic and kinetic vehicle models with steerable tractor and dolly axles. The kinetic model also accounts for propulsion forces. Using these models, we set up an optimization problem with the CasADi toolbox for MATLAB, focusing on optimizing actuator inputs to enhance vehicle performance during various maneuvers.

1.3 Limitations of scope

In the course of this thesis, several limitations had to be taken into consideration for the sake of reducing the problem complexity and keeping the time frame of the project. These limitations have an impact on the overall optimization and real-time application of the proposed method for LCVs. These limitations are detailed below:

- Both kinematic and kinetic models are single-track bicycle models (Wheels on left and right are combined as a single wheel on axle midpoint).

- The kinematic model is valid for steady state conditions.
- The kinematic model assumes zero lateral velocity at the wheels i.e. zero side slip at the wheels.
- The kinematic model is more accurate for low speed maneuvers.
- In the kinetic model the tire model is considered to be linear.
- There is no longitudinal slip taken into consideration in the kinetic model.
- Optimization is restricted to minimizing swept path and yaw amplification.
- Most maneuvers used to test the model are low speed maneuvers almost attending steady state, except lane change.
- The optimization model has a short coming of real time application on any high fidelity vehicle model due to time constraint.

1.4 Research Questions

Here are the research questions that this thesis focuses on, which we aim to address by the conclusion of this work::

- What is the optimal control strategy to enhance maneuverability and lateral stability of LCVs in challenging situations?
- Which vehicle model performs best within this control strategy?
- How effective and robust are the cost functions used in optimal control for achieving the desired objectives in specific maneuvers?

1.5 Outline of Report

In the following pages, this report will delve into the theoretical understanding underpinning our study, providing a comprehensive overview of the key concepts and knowledge that form the basis of our work. This foundational theory will include a detailed examination of vehicle dynamics, the principles of the bicycle model. Following the theoretical framework, we will describe the methodology employed for building the models and optimizing the dolly steering system. The process of developing the kinematic and kinetic models using the bicycle model approach. We will outline the steps taken to parameterize the models based on realistic vehicle data. A step-by-step description of the optimization process, including the design of the control algorithm, the application of optimization techniques, and the specific constraints considered. An overview of the simulation environment, including the software tools used (such as MATLAB) and the setup for virtual testing. The scenarios and driving maneuvers tested in the simulations will be detailed. The results section then will present the findings of our study, using various data visualizations to highlight key insights. A detailed evaluation of the optimized dolly steering system using PBSs. Key metrics such as swept path width. A comparison of the performance of the kinematic and kinetic models, discussing their respective strengths and weaknesses. Finally, we will conclude by summarizing the main findings of our study, discussing their implications, acknowledging any limitations, and offering recommendations for future research and practical applications

2

Vehicle Dynamics

To set up an optimization scenario for optimal control allocation of an A-double combination, an insight into the mathematical modelling and vehicle dynamics is needed to understand and take into consideration the impact of system dynamics on achieving a desired objective.

All the modelling process is carried out as per the standard coordinate system used for vehicle frame. This is a right hand coordinate system, fixed in the center of gravity of the vehicle and is originated from the definition in ISO 8855.

The models described in the following sections is only based on planar dynamics. This means that the only interesting motions are the longitudinal, lateral and yaw motion. The longitudinal motion is defined along the x-axis, the lateral motions along the y-axis, and the rotation around the z-axis is referred to as the yaw motion.

In addition to the model being restricted to the planar dynamics, resistive forces like aerodynamic and rolling resistances are neglected, the units are considered as rigid masses and the left and right wheels have equal steer angle. There are also several assumptions regarding the tire dynamics which are discussed later.

Two models were derived for the combination, the first one is a more simplistic kinematic model based on geometric relations and velocity and the second one a more complex kinetic model, which take into account forces acting on the combination. Both vehicle models are single-track bicycle models, where the left and right wheels of an axle are merged and represented as a single wheel. For the kinetic models, linear tire models are presented.

2.1 Vehicle Configuration

In this thesis, the combination used is an A-double with steerable e-dolly. As the name suggests this is a combination with double trailers with a tractor and e-dolly. The dolly, which is used to connect the second trailer with the first trailer is also considered to have a steerable and propelled axles. To simplify the modelling and optimisation, the axles on the every unit are lumped. Hence in the vehicle model being investigated, the combination has a total of 5 axles. Two axles for the tractor and one axle each for the two trailers and e-dolly.
The dimensions of the combination are preserved as it is, each trailer has a length of

40-foot which makes the total combination length to be 32.5 meters. The considered track width for the entire combination is 2.5 meters.

2.2 Kinematic Bicycle Model

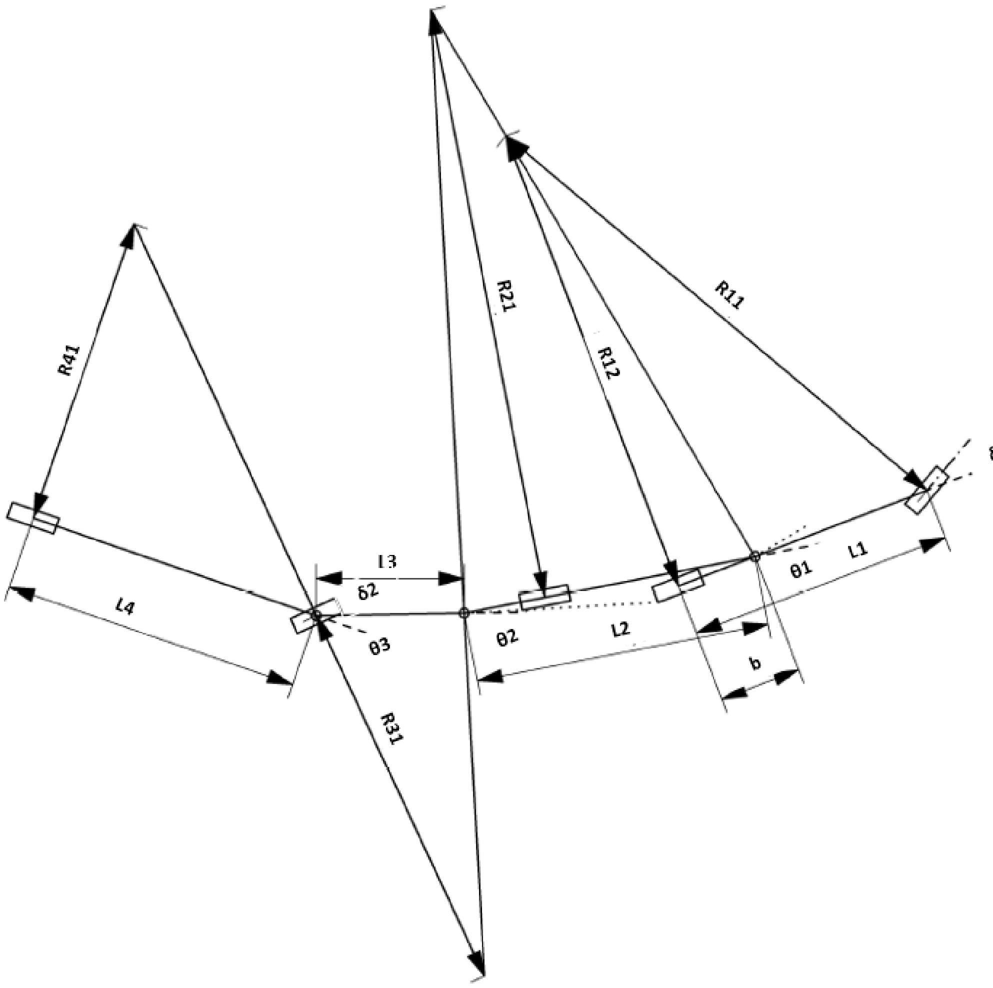


Figure 2.1: Kinematic model for A-double with steerable dolly.

The kinematic bicycle model is one of the simpler models of an A-double combination. This model is derived by following the approach of instantaneous centers, As shown in Figure 2.1 the motion of every unit of the combination can be broken down into rotation around an instantaneous center. These instantaneous centers are formed when perpendicular lines drawn from the velocity vectors of the axles intersect [4]. It is used to understand the geometric relations between several units of the vehicle in motion and to further simplify, an assumption is made that the center of gravity (CoG) of each unit is located at the midpoint of it's rear axle. Based on these geometric relations the equations for turning radius of every axle was derived as below.

The turning radius of the tractor's front axle is given by:

$$R_{11} = \sqrt{(R_{12}^2 + L_1^2)} \tag{2.1}$$

The turning radius of the tractor's rear axle is given by:

$$R_{12} = \frac{L_1}{\tan(\delta_1)} \tag{2.2}$$

The turning radius of first trailer's rear axle is given by:

$$R_{21} = \frac{L_2}{tan(\theta_1 + \beta_1)} \tag{2.3}$$

The turning radius of the dolly's axle is given by:

$$R_{31} = \frac{L_3(\cos(\beta_2 + \theta_2)}{sin(-\delta_2 + \beta_2 + \theta_2)} \tag{2.4}$$

The turning radius of the second trailer's rear axle is given by:

$$R_{41} = \frac{L_4}{\tan(\delta_2 + \theta_3)} \tag{2.5}$$

Further, in order use these equations in the optimisation problem, it is necessary to have time dependent equations. Since the radius of a unit can also be written as a ratio of longitudinal velocity to the yaw rate of that unit, the time dependent kinematic equations were obtained as below:

$$R_{ij} = \frac{V_{Xi}}{\phi_i}, \tag{2.6}$$

where R_{ij} is the turning radius of the jth axle of the ith unit, V_{xi} is the longitudinal velocity of that unit and ϕ_i is the yaw rate.

This approach can be extended to n number of units with n steerable axles. With the addition of steerable axle, the geometric relations change and the kinematic model needs to be reworked following the same principle of instantaneous centers.

The kinematic model has certain limitations. Firstly, it is only applicable for low-speed, steady-state maneuvers. Secondly, due to the low speed, the slip angle at the wheels is negligible. Lastly, this model does not take into account any forces acting on the vehicle.

2.3 Kinetic Bicycle Model

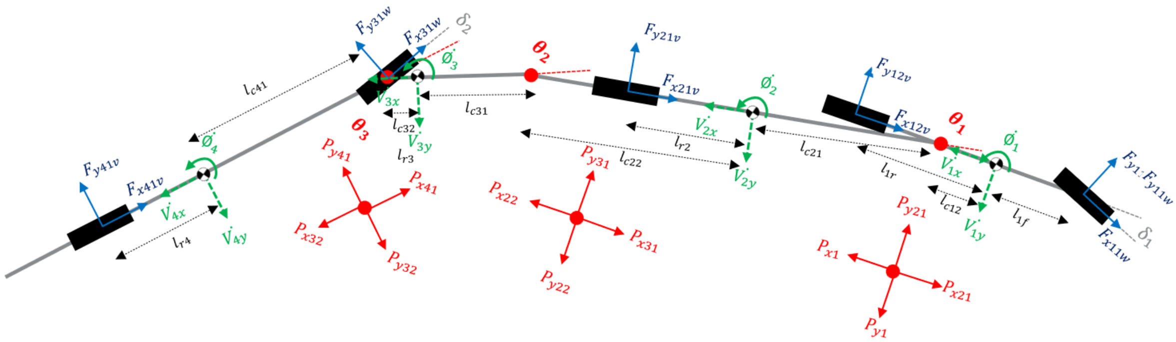


Figure 2.2: Kinetic bicycle model for A-double with steerable dolly.

The kinematic model is valid only for low-speed maneuvers. As speed increases, the model's accuracy diminishes due to the need to account for dynamic effects such as slip angles and forces acting on the vehicle combination. To address this, a single-track kinetic model was developed [5], as illustrated in Figure 2.2.

The equations of motion for this bicycle model are formulated using the Newtonian approach, which involves setting up force and moment balance equations for each unit in the combination. Since this model focuses solely on planar dynamics, roll and pitch motions are neglected. Consequently, each unit has three equations governing lateral force balance, longitudinal force balance, and moment balance along the z-axis. These equations can be solved using MATLAB's Symbolic Toolbox to obtain the desired explicit or implicit equations for the variables of interest in the optimization problem.

Below is the list of equations for the A-double combinations.

Equilibrium for tractor:

$$m_1\ (\dot{v}_{1x} - \dot{\phi}_1\ v_{1y}) = F_{x11w}\ cos(\delta_1) - F_{y11w}\ sin(\delta_1) + F_{x12v} - P_{x1} + P_{x21} \tag{2.7}$$

$$m_1\ (\dot{v}_{1y} + \dot{\phi}_1\ v_{1x}) = F_{x11w}\ sin(\delta_1) + F_{y11w}\ cos(\delta_1) + F_{y12v} - P_{y1} + P_{y21} \tag{2.8}$$

$$J_1\ \ddot{\phi}_1 = (sin(\delta_1)\ F_{x11w} + cos(\delta_1)\ F_{y11w})\ l_{1f} - F_{y12v}\ l_{1r} + P_{y1}\ l_{c12} - P_{y21}\ l_{c12} \tag{2.9}$$

Equilibrium for first trailer:

$$m_2\ (\dot{v}_{2x} - \dot{\phi}_2\ v_{2y}) = F_{x21v} + P_{x31} + P_{x21} - P_{x22} \tag{2.10}$$

$$m_2\ (\dot{v}_{2y} + \dot{\phi}_2\ v_{2x}) = F_{y21v} + P_{y31} + P_{y21} - P_{y22} \tag{2.11}$$

$$J_2\ \ddot{\phi}_2 = -F_{y21v}\ l_{2r} + P_{y22}\ l_{lc22} - P_{y31}\ l_{lc22} + P_{y21}\ l_{lc21} \tag{2.12}$$

Equilibrium for dolly:

$$m_3\ (\dot{v}_{3x} - \dot{\phi}_3\ v_{3y}) = F_{x31w}\ cos(\delta_2) - F_{y31w}\ sin(\delta_2) - P_{x41} + P_{x31} - P_{x32} \tag{2.13}$$

$$m_3\ (\dot{v}_{3y} + \dot{\phi}_3\ v_{3x}) = F_{x31w}\ sin(\delta_2) + F_{y31w}\ cos(\delta_2) - P_{y32} + P_{y41} + P_{y31} \tag{2.14}$$

$$J_3\ \ddot{\phi}_3 = -(sin(\delta_2)\ F_{x31w} + cos(\delta_2)F_{y31w})\ l_{3f} - P_{y41}\ l_{c32} + P_{y32}\ l_{c32} + P_{y31}\ l_{c31} \tag{2.15}$$

Equilibrium for second trailer:

$$m_4\ (\dot{v}_{4x} - \dot{\phi}_4\ v_{4y}) = F_{x41v} + P_{x41} - P_{x32} \tag{2.16}$$

$$m_4\ (\dot{v}_{4y} + \dot{\phi}_4\ v_{4x}) = F_{y41v} + P_{y41} - P_{y32} \tag{2.17}$$

$$J_4\ \ddot{\phi}_4 = -F_{y41v}\ l_{4r} + P_{y41}\ l_{lc41} - P_{y32}\ l_{lc41} \tag{2.18}$$

Coupling forces relations - coupling 1:

$$P_{x21} = -(P_{x1}\ cos(\theta_1) - P_{y1}\ sin(\theta_1)) \tag{2.19}$$

$$P_{y21} = -(P_{x1}\ sin(\theta_1) + P_{y1}\ sin(\theta_1)) \tag{2.20}$$

Coupling forces relations - coupling 2:

$$P_{x31} = -(P_{x22}\ cos(\theta_2) - P_{y22}\ sin(\theta_2)) \tag{2.21}$$

$$P_{y31} = -(P_{x22}\ sin(\theta_2) + P_{y22}\ sin(\theta_2)) \tag{2.22}$$

Coupling forces relations - coupling 3:

$$P_{x41} = -(P_{x32}\ cos(\theta_3) - P_{y32}\ sin(\theta_3)) \tag{2.23}$$

$$P_{y41} = -(P_{x32}\ sin(\theta_3) + P_{y32}\ sin(\theta_3)) \tag{2.24}$$

2.3.1 Tyre Model

A simplified tire model is employed for the above vehicle model [5], assuming the vehicle has a longitudinal slip-control onboard, it is not explicitly considered in the equations. For the lateral characteristic, the cornering stiffness is required to calculate the about of lateral force generated for a certain slip angle.

By using a fixed average value for cornering stiffness of tyre on either side of the axle, we assume a linear relationship between slip angle and lateral force. This assumption is effective for the small slip angles relevant to this study.

Linear tyre model for tractor:

$$s_{11y} = atan(v_{1yfw}/v_{1xfw}) \qquad s_{12y} = atan(v_{1yrv}/v_{1xrv}) \tag{2.25}$$

$$F_{y11w} = -C_{1f}\ s_{11y} \qquad F_{y12v} = -C_{1r}\ s_{12y} \tag{2.26}$$

Linear tyre model for first trailer:

$$s_{21y} = atan(v_{2yv}/v_{2xv}) \tag{2.27}$$

$$F_{y21v} = -C_{2r}\ s_{21y} \tag{2.28}$$

Linear tyre model for dolly:

$$s_{31y} = atan(v_{3yv}/v_{3xv}) \tag{2.29}$$

$$F_{y31v} = -C_{3r}\ s_{31y} \tag{2.30}$$

Linear tyre model for second trailer:

$$s_{41y} = atan(v_{4yv}/v_{4xv}) \tag{2.31}$$

$$F_{y41v} = -C_{4r}\ s_{41y} \tag{2.32}$$

There has to be compatibility equations which represents the velocity of other units with respect to tractor's as shown below the relations:

Velocity of tractor in vehicle-frame:

$$\begin{aligned} v_{1xfv} &= v_{1x} \\ v_{1yfv} &= v_{1y} + l_{1f}\ \dot{\phi}_1 \end{aligned} \tag{2.33}$$

$$\begin{aligned} v_{1xrv} &= v_{1x} \\ v_{1yrv} &= v_{1y} - l_{1r}\ \dot{\phi}_1 \end{aligned} \tag{2.34}$$

Velocity of tractor in wheel-frame:

$$\begin{aligned} v_{1xfw} &= v_{1xfv}\ cos(\delta_1) + v_{1yfv}\ sin(\delta_1) \\ v_{1yfw} &= v_{1yfv}\ cos(\delta_1) - v_{1xfv}\ sin(\delta_1) \end{aligned} \tag{2.35}$$

Velocity relation for coupling 01 (On-Tractor):

$$\begin{aligned} v_{1xc} &= v_{1xv} \\ v_{1yc} &= v_{1yv} - l_{c12}\ \dot{\phi}_1 \end{aligned} \tag{2.36}$$

Velocity relation for coupling 01 (On-Trailer):

$$\begin{aligned} v_{21xc} &= v_{1xc}\ cos(\theta_1) - v_{1yc}\ sin(\theta_1) \\ v_{21yc} &= v_{1yc}\ cos(\theta_1) + v_{1xc}\ sin(\theta_1) \end{aligned} \tag{2.37}$$

Velocity relation for first trailer CoG:

$$\begin{aligned} v_{2xv} &= v_{21xc} \\ v_{2yv} &= v_{21yc} - (l_{21c} + l_{2r})\dot{\phi}_2 \end{aligned} \tag{2.38}$$

Velocity relation for coupling 02 (on-trailer):

$$\begin{aligned} v_{22xc} &= v_{2xv} \\ v_{22yc} &= v_{2yv} - (l_{22c})\ \dot{\phi}_2 \end{aligned} \tag{2.39}$$

Velocity relation for coupling 02 (on-dolly):

$$\begin{aligned} v_{31xc} &= v_{22xc}\ cos(\theta_2) - v_{22yc}\ sin(\theta_2) \\ v_{31yc} &= v_{22yc}\ cos(\theta_2) + v_{22xc}\ sin(\theta_2) \end{aligned} \tag{2.40}$$

Velocity relation for dolly CoG:

$$\begin{aligned} v_{3xv} &= v_{31xc} \\ v_{3yv} &= v_{31yc} - (l_{31c} + l_{3r})\ \dot{\phi}_3 \end{aligned} \tag{2.41}$$

Velocity relation for coupling 03 (on-dolly):

$$\begin{aligned} v_{32xc} &= v_{3xv} \\ v_{32yc} &= v_{3yv} - (l_{32c})\ \dot{\phi}_3 \end{aligned} \tag{2.42}$$

Velocity relation for coupling 03 (on-trailer):

$$\begin{aligned} v_{41xc} &= v_{32xc}\ cos(\theta_3) - v_{32yc}\ sin(\theta_3) \\ v_{41yc} &= v_{32yc}\ cos(\theta_3) + v_{32xc}\ sin(\theta_3) \end{aligned} \tag{2.43}$$

Velocity relation for second trailer CoG:

$$\begin{aligned} v_{4xv} &= v_{41xc} \\ v_{4yv} &= v_{41yc} - (l_{41c} + l_{4r})\ \dot{\phi}_4 \end{aligned} \tag{2.44}$$

2.4 Vehicle Models Validation

In order to verify the above two models Volvo high fidelity model, Volvo Transport Model (VTM), was used. The methodology for which is quite simple convert both the models into state space form with carefully selecting states and input vector. Take the input and necessary states from the VTM, calculate the unknown states and their derivative and finally compare. For this, maneuver was circle of fixed radius of 25m, with vehicle running at constant low speed and road friction of 0.8, in order to achieve steady state.

2.4.1 Verification for Kinematic Model

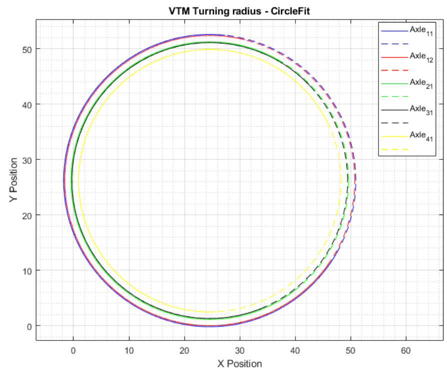
A time based model was obtained using kinematic equations. Vehicle parameters and some sensor data like articulation angles, steering angle and tractor speed along the path (5m/sec)was taken from the VTM. These were used to calculate position, yaw rates and, longitudinal and lateral velocity of each units along the path.

The comparison below highlights the results from the VTM and the kinematic model for both steered and non-steered dolly in A-double combination. The plots in Figure 2.3 illustrate this comparison: the vehicle trajectory and the yaw rates of the individual units. The dotted line in the plots represents the actual data from the VTM, while the solid line corresponds to the kinematic model. The close resemblance between the two is striking, the maximum deviation in yaw rate that was noted in any unit of the combination was less than 3%. The reason for this similarity is straightforward: the kinematic model uses inputs derived from the VTM, and the maneuver is conducted at a steady state with very low vehicle velocity. If the speed is increased the effect of inertia and forces come into picture ruining the similarities.

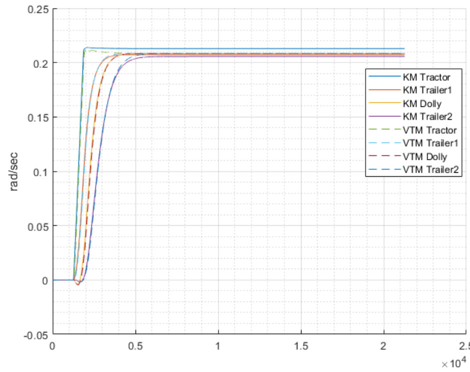
2.4.2 Verification for Kinetic Model

The single track model described above needed verification against the VTM. To achieve this, similar sensor data and vehicle parameters were logged from the VTM too. The maneuver used for this verification was the same as the one used for the kinematic model verification, except for the radius of curvature, which was set to 30m in this case. Verifying the kinetic model required only a vehicle combination with a non-steered dolly, where an in-build path follower of VTM was used to perform a maneuver at constant speed (10m/sec) and constant steering angle. Subsequently, similar inputs was used for verification.

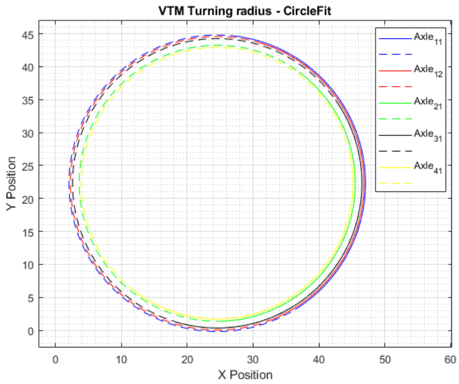
The comparison is illustrated through three different plots. Figure 2.4b demonstrates the kinetic model's ability to account for forces, showing results that closely align with the VTM. The other two plots, Figures 2.4a and 2.4c, highlight the strong performance of the single track model in comparison to the VTM during simpler maneuvers



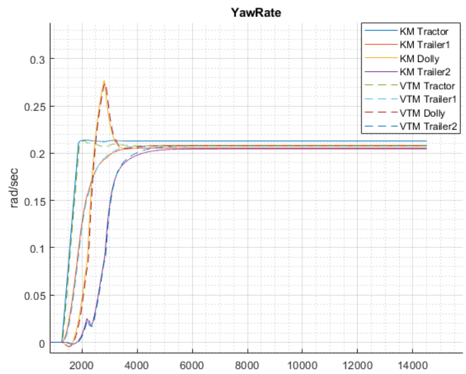
(a) Vehicle trajectory for non-steered dolly



(b) Yaw rate of individual units for non-steered dolly



(c) Vehicle trajectory for steered dolly

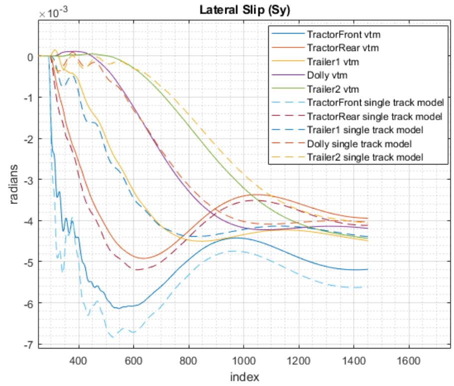


(d) Yaw rate of individual units steered dolly

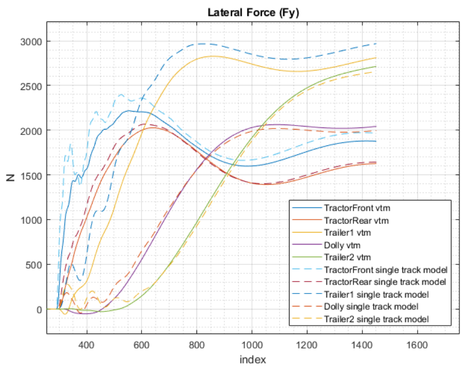
Figure 2.3: Comparison between VTM and kinematic model for both non-steered and steered dolly in A-double combination

2.5 Performance Based Characteristics

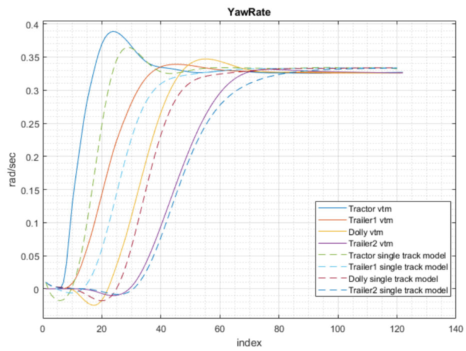
The main criterion for determining if a vehicle should be allowed on the road is its performance and interaction with the road network. Performance-Based Characteristics (PBC) is one standard that assesses vehicle behavior both longitudinally and laterally. PBC evaluates how effectively and safely vehicles navigate various driving situations, such as turns, lane changes, and tight spaces [6]. LCVs, typically a truck tractor pulling multiple trailers, have increased overall length, impacting maneuverability, stability, and safety for both the vehicle and its surroundings. This thesis focuses specifically on the lateral performance of an A-double combination.



(a) Side slip angle at tyre



(b) Lateral force acting on the tyre



(c) Yaw rates of individual units

Figure 2.4: Comparison between VTM and Single track model for a non-steered dolly in A-double combination

2.5.1 Swept Path Width

The width of the swept path of a vehicle is a crucial parameter for comparing the performance of different articulated vehicle configurations. It measures the vehicle's ability to navigate roundabouts by indicating the maximum radial distance between the inner and outermost points of the vehicle's path when traveling around a curvature at low speed [5], as shown in Figure 2.5. This metric helps assess how well the vehicle can maneuver in tight, circular spaces, such as roundabouts.